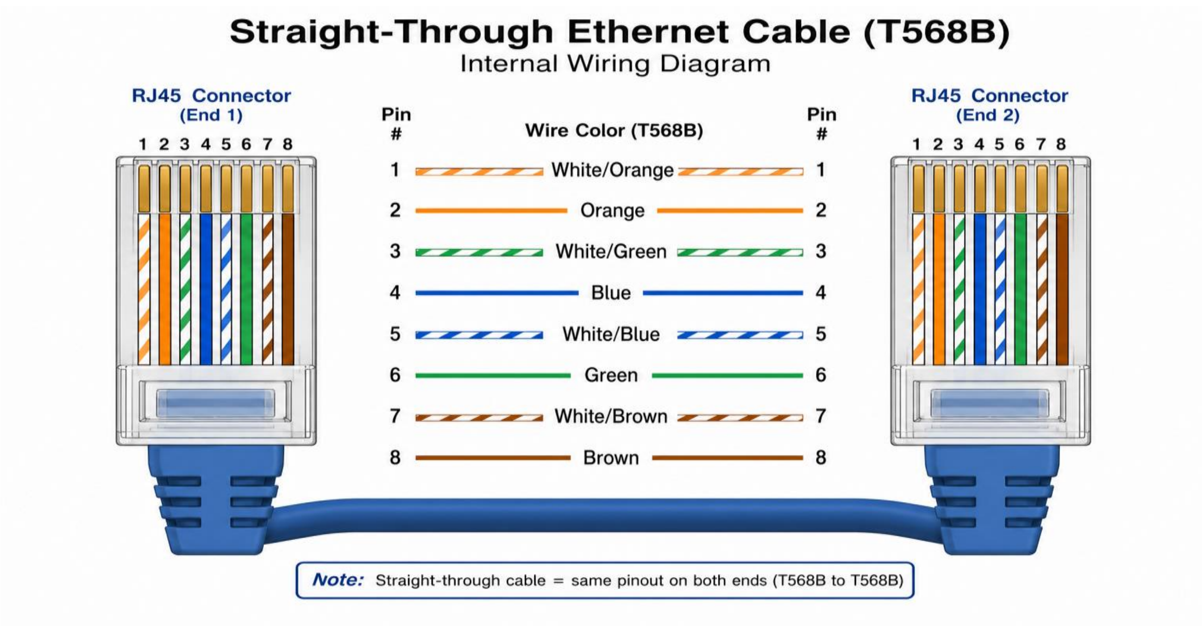
1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home Wi-Fi router, data center, college campus backbone).

Feature	Ethernet Cable	Fiber Optic Cable
Transmission Medium	Sends data using electrical signals through copper wires.	Sends data using pulses of light through glass or plastic fibers.
Speed and Distance	Typically supports high speeds over shorter distances.	Supports much higher speed over longer distances.
Interference	Susceptible to electromagnetic interference from nearby electrical equipment.	Immune to electromagnetic interference, making it more reliable in electrically noisy environments.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

- I reviewed a YouTube demonstration of crimping and RJ45 connector onto a CAT6 cable and used it as the basis for the guide as follows:

1. **Cut the cable**

- Cut the CAT6 cable to the required length using the cutter on the crimping tool or a pair of cable cutters.

2. **Strip the outer jacket**

- Remove about 2-3cm of the outer jacket from the end of the cable. Be careful not to nick or damage the insulated wires inside.

3. **Prepare the wires**

- Separate the four twisted pairs and untwist only as much as needed. Straighten each wire with your fingers to make them easier to arrange.

4. **Arrange the wires**

- Place the wires in the correct order. The most common standard is T568B:
- White/Orange
- Orange
- White/ Green
- Blue
- White/Blue
- Green
- White/Brown
- Brown

5. **Trim the wires evenly**

- Hold the wires tightly together and trim their ends so they are all the same length, leaving about 12-13mm extending from the cable jacket.

6. **Insert the wires into the RJ45 connector**

- With the connector’s clip facing downward; carefully slide the wires into the connector while keeping them in the correct order. Push them fully forward until each wire reaches the front of the connector, and ensure the cable jacket extends slightly into the connector for strain relief.

7. **Crimp the connector**

- Insert the RJ45 connector into the crimping tool and squeeze the handles firmly until the crimp is complete. This pushes the metal contacts into the wires and secures the connector onto the cable.

8. **Inspect the connection**

- Check that all eight wires remain in the correct order, reach the end of the connector, and that the jacket is firmly held by the connector's strain-relief tab.

9. **Test the cable**

- Use a network cable tester to verify that all eight conductors are connected correctly and that there are no wiring faults before putting the cable into service.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

- For a gaming cafe with 10 PCs, I would use Ethernet cables because they provide low latency, high-speed connections, are reliable for LAN gaming, and are much more cost-effective than fiber for short indoor distances. I would use an RJ45 crimper to attach RJ45 connectors to the Ethernet cables. Fiber optics and a fiber cleaver are better suited for long-distance or backbone network connections, but they would be unnecessarily expensive and complex for connecting just 10 PCs in the same room.